Hostel Management System

Code Coverage Report

Statement coverage achieved by the automated test suite

Project	Hostel Management System (HMS)
Team	Team 3
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Repository	github.com/adarshv15/hostel-management-system-team3 (latest tested working tree; commit not re-verified)
Baseline document	Software Requirements Specification v1.0 (TEAM_3_HMS_SRS)
Report version / date	1.1 / 10-11-2026
Status	Draft for team review
Test period	Latest local coverage run; 38-test suite

1. Introduction

This report gives the statement (line) coverage achieved by the automated test suite on the Hostel Management System (HMS). Statement coverage identifies executable application statements that ran at least once while the automated tests ran. It does not show whether the tests check the right results; that is covered by the Unit Test Report.

Branch coverage, which measures whether both outcomes of each decision were taken, is reported separately in the Branch Coverage Report.

2. Measurement Method

Item	Detail
Operating system	Windows 10/11 (local development machine; exact build not recorded)
Python	3.14.0
Django	5.2.19 (previously recorded; local version not re-verified)
Coverage tool	coverage.py 7.16.2 with --branch
Database	SQLite, Django-managed test database
Command	python -m coverage run --branch --source=hostel,hms_project manage.py test (38-test suite; latest test run)
Reports	python -m coverage report -m --omit="*/tests.py"; python -m coverage json --omit="*/tests.py" -o coverage_app.json
Scope	Application Python files in hostel/ and hms_project/. The test module hostel/tests.py is excluded so test code does not inflate application coverage. Empty __init__.py files have no executable statements.
Execution date	10-11-2026

3. Summary of Results

Measure	Result
Executable statements	324
Statements executed	299
Statements missed	25
Statement coverage	92.3% (299 of 324)
Test result	38 of 38 tests passed (100%)

Result: statement coverage is 92.3% (299 of 324 executable statements). The two main application files, models.py (146 of 155 statements; 94.2%) and admin.py (83 of 91; 91.2%), both exceed 90%. The signals.py module is now fully covered (17 of 17 statements), and all 26 application branches were taken. The remaining 25 missed statements are concentrated in admin.py (8), models.py (9), asgi.py (4) and wsgi.py (4).

The Test Plan's coverage target was not available in the repository. If it specifies a threshold, compare it with the 92.3% statement-coverage figure above.

3.1 Coverage by file

File	Statements	Executed	Missed	Coverage
hostel/models.py	155	146	9	94.2%
hostel/admin.py	91	83	8	91.2%
hostel/signals.py	17	17	0	100.0%
hostel/views.py	8	8	0	100.0%
hostel/apps.py	7	7	0	100.0%
hostel/migrations/0001_initial.py	9	9	0	100.0%
hms_project/settings.py	25	25	0	100.0%
hms_project/urls.py	4	4	0	100.0%
hms_project/asgi.py	4	0	4	0.0%
hms_project/wsgi.py	4	0	4	0.0%
Total	324	299	25	92.3%

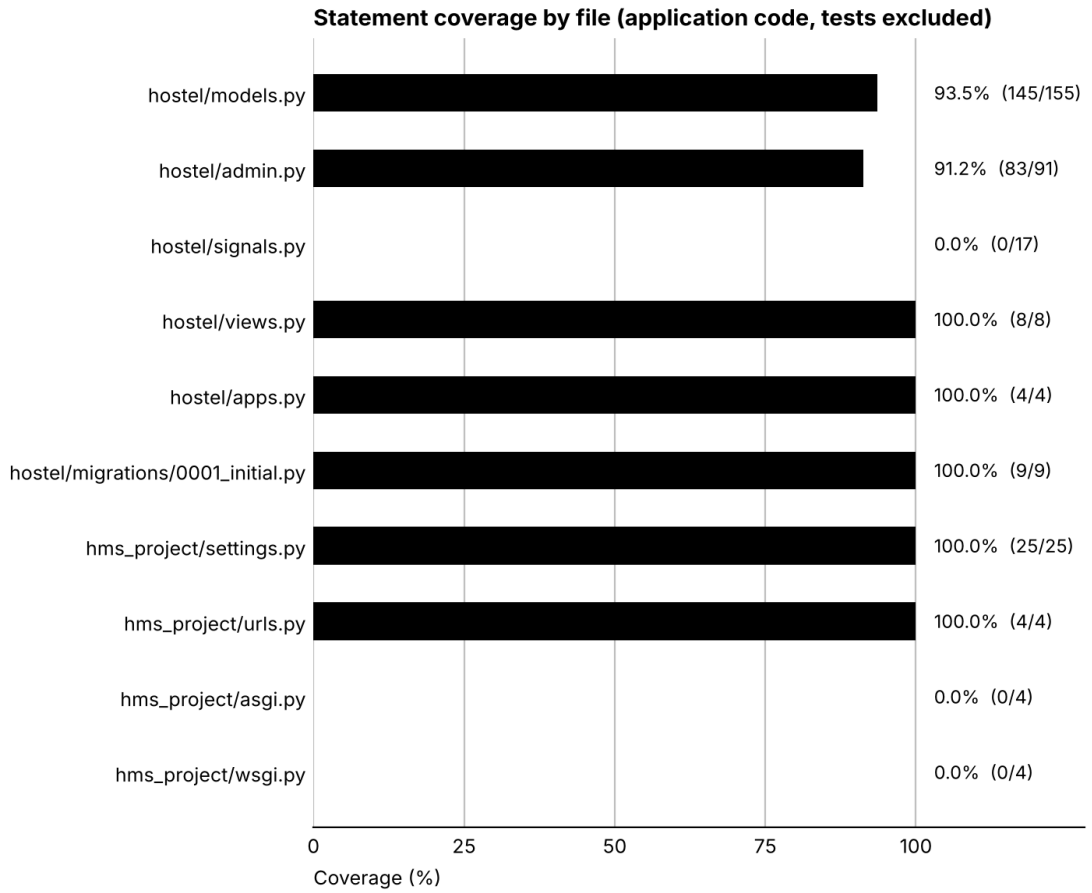


Figure 1. Statement coverage for each application file.

3.2 Sensitivity to scope

Scope	Statements	Missed	Coverage
Application code, tests excluded (headline)	324	25	92.3%
Same, also excluding asgi.py and wsgi.py (entry points)	316	17	94.6%

The headline figure excludes hostel/tests.py and includes all application files with executable statements. Excluding the ASGI and WSGI deployment entry points as well gives 94.6% (299 of 316 statements).

4. Uncovered Code

All 25 missed application statements, grouped by file and likely cause.

Location	Lines	Code	Risk	Cause
hostel/admin.py	45-58	Single-record delete_model audit event and deletion path	Low	Single-record admin deletion is not exercised
hostel/admin.py	152-159, 169, 173-174	FeeInvoiceAdmin annotated queryset and paid/balance display columns	Medium	Invoice admin changelist/display columns are not rendered in tests
hostel/models.py	71, 84, 115, 130, 149, 166, 180, 191, 206	Model __str__ methods for the listed domain models	Low	String representations are not asserted by tests

Location	Lines	Code	Risk	Cause
hms_project/asgi.py	1-4	ASGI application entry point	Low	Not invoked by Django's test client
hms_project/wsgi.py	1-4	WSGI application entry point	Low	Not invoked by Django's test client

Missed statements by file: admin.py 8, models.py 9, asgi.py 4 and wsgi.py 4. Total: 25. signals.py has improved from 0% to 100% statement coverage.

5. Recommendations

The remaining gaps can be addressed with a small set of focused tests:

1. Exercise FeeInvoiceAdmin's changelist after recording a part payment and assert the displayed paid and balance values (covers the remaining invoice-admin display statements).
2. Delete one record through the admin and assert that the expected DELETE audit event is created (covers the single-record delete path in admin.py).
3. Assert the string representation of each model whose __str__ statement is currently uncovered (covers the nine missed lines in models.py).
4. ASGI and WSGI entry points are deployment interfaces rather than ordinary application business logic. If they are in the required coverage scope, add focused import/initialization checks; otherwise document their exclusion consistently.
5.

The current application statement coverage is 92.3% (299 of 324). Covering the 17 remaining statements in admin.py and models.py would raise it to 97.5% (316 of 324). Covering the four statements in each deployment entry point would bring the measured application scope to 100%.

6. Limitations

- Statement coverage shows that a line ran, not that its result was checked. A covered line may be executed without every output or side effect being asserted.
- Coverage does not measure requirements that are not implemented. Functional completeness and requirement validation are addressed separately in the System Validation Report.
- Figures come from a local Windows run using Python 3.14.0 and coverage.py 7.16.2. The local Django version was not re-verified for this run, and results on the project's CI environment were not measured.
- Template rendering and Django internals are not measured.

7. How to Reproduce

```
python -m pip install -r requirements.txt
python -m coverage erase
python -m coverage run --branch --source=hostel,hms_project manage.py test
python -m coverage report -m --omit="*/tests.py"
python -m coverage json --omit="*/tests.py" -o coverage_app.json
```

8. Conclusion

The latest 38-test suite achieves 92.3% statement coverage across the measured application code, with all 26 branches covered (100% branch coverage). Signal registration and execution are now covered. The

remaining statement gaps are concentrated in admin display/deletion paths, model string representations, and the ASGI/WSGI entry points.

Sign-off

Role	Name	Signature	Date
Prepared by			
Reviewed by			
Approved by (Project Guide)			

Appendix A. Raw coverage.py Output

Output of coverage report -m --omit="*/tests.py" from the latest local run. The Cover column combines statement and branch coverage when --branch is enabled; the report headline above uses statement coverage only.

```
Name                                  Stmts   Miss Branch BrPart  Cover   Missing
-----------------------------------------------------------------------------------
hms_project/asgi.py                       4      4      0      0     0%   1-4
hms_project/settings.py                  25      0      0      0   100%
hms_project/urls.py                       4      0      0      0   100%
hms_project/wsgi.py                       4      4      0      0     0%   1-4
hostel/admin.py                          91      8      2      0    91%   45-58, 152-159, 169, 173-
174
hostel/apps.py                            7      0      0      0   100%
hostel/migrations/0001_initial.py         9      0      0      0   100%
hostel/models.py                        155      9     20      0    95%   71, 84, 115, 130, 149,
166, 180, 191, 206
hostel/signals.py                        17      0      4      0   100%
hostel/views.py                           8      0      0      0   100%
-----------------------------------------------------------------------------------
TOTAL                                   324     25     26      0    93%


3 empty files skipped.
```